

Forecasting French Day-Ahead Power Prices

Weather- and fundamentals-driven ML model for the EPEX Spot FR auction · Léo Marie ·

github.com/leomariecode/EnergeticPricing

Target	Information set	Backtest	Main model	vs. naive D-1
FR day-ahead price, 24 hourly values (EUR/MWh)	Only data known before gate closure (D-1, 12:00 CET)	Jul 2023 – May 2026, 5 expanding folds	LightGBM, 504 features	MAE –26% (22.3 → 16.5) once ≥ 18 months of history

1 · Problem

The French day-ahead price is set every day at 12:00 CET by the coupled European auction (EPEX Spot / EUPHEMIA) for each hour of the next day (15-minute products since Oct. 2025). It is the reference for spot hedging, battery and flexibility scheduling, and the anchor of intraday trading. It is driven by **residual load** (demand minus wind and solar), by the **marginal technology** (gas plants set the price when nuclear, hydro and renewables do not cover demand) and by **cross-border coupling**. The project forecasts the 24 hourly prices of D+1 using **only information available before gate closure**, and measures what each block of information is worth against naive persistence.

2 · Data

Source	Variables	Granularity	Use in the model
ENTSO-E Transparency Platform	FR day-ahead price (target); day-ahead forecasts of load, wind and solar; DA prices of DE, BE, ES, IT-North, CH	Hourly (15 min since Oct. 2025)	Target; residual load and its ramps. Neighbour prices only in the <i>reconstruction</i> upper bound
Open-Meteo (ERA5 archive + forecast API)	Temperature, 10 m wind speed, shortwave radiation, cloud cover, precipitation	Hourly, 96 départements (prefecture coordinates)	480 spatial weather features (5 variables × 96 départements)
ODRE national installation registry	Solar and wind capacity per département with commissioning date	Daily steps	Point-in-time installed capacity; <i>renewable potential</i> = $\sum_{\text{dept}^3} \text{capacity} \times \text{radiation (solar)}$ or $\times \text{wind speed}^3$ (wind)
ICE Endex TTF front-month (via Yahoo Finance)	Natural-gas price (EUR/MWh)	Daily close	Marginal-cost proxy, lagged 2 days (last settlement known at gate closure)
ENTSO-E · nuclear	Actual nuclear output; planned unavailability messages (REMIT)	Hourly · per event	Output of D-2, planned outage MW at D, their change vs D-1, residual load net of available nuclear (tested, not retained)
INSEE · French calendar	Population by département · public holidays	Static · daily	Population-weighted national weather; holidays treated as weekends

Sample: Jan 2023 – Apr 2026, about 29,200 hourly observations. All API calls are cached as parquet; the dataset and every result in this note rebuild with one command.

3 · Methodology

- **No look-ahead.** Each feature is checked against what is known at D-1 noon: price lags at D-1 and D-7, gas settlement of D-2, nuclear output of D-2, ENTSO-E day-ahead forecasts, calendar. Neighbour DA prices clear *simultaneously* with France, so they are excluded from all forecasting models and kept only in a *reconstruction* run that shows how large this leak is.
- **Features from market structure:** residual-load forecast and hour-to-hour ramps (morning and evening solar ramps), capacity-weighted renewable potential, cyclical hour and month encodings, French public holidays.
- **Validation:** rolling-origin cross-validation with 5 expanding windows of about 6.5 months of test each. LightGBM early-stops on the tail of the *training* window, never on the test fold.
- **Benchmarks:** naive persistence (same hour D-1 and D-7) on the same folds. Ridge regression is the linear baseline, LightGBM the main model, and an auxiliary LightGBM classifier flags price spikes (outside the 5–95% quantiles, including negative prices).

4 · Results

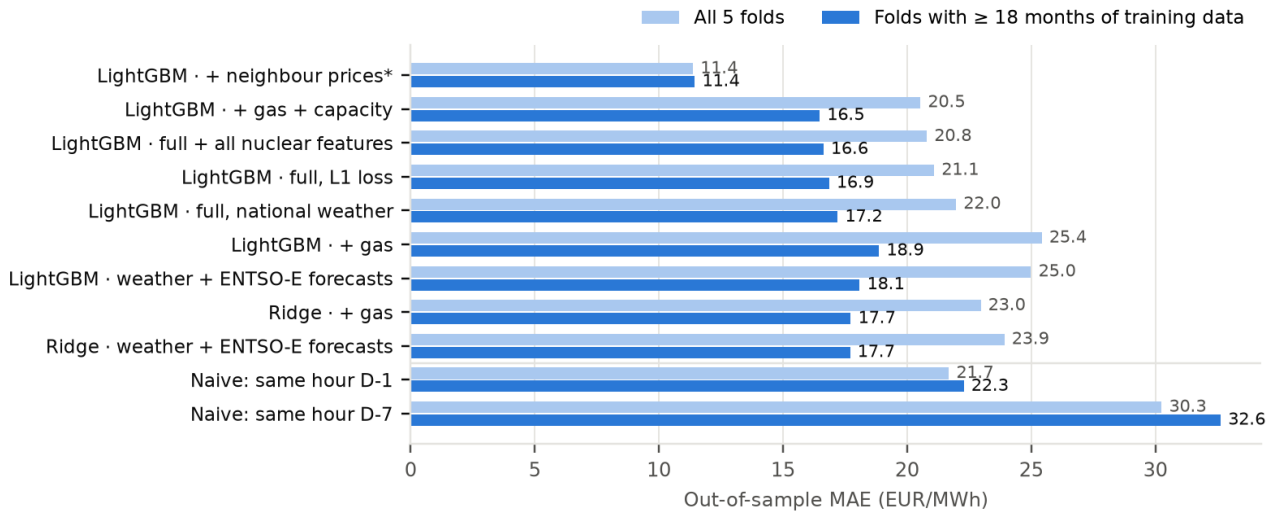


Figure 1 – Out-of-sample MAE by model and feature set (lower is better). Dark bars: the 3 folds tested from Aug 2024 onwards, trained on ≥ 18 months. *Uses neighbour prices that are unknown before the auction: an upper bound, not a forecast.

Model · features	MAE, all folds	MAE, ≥ 18 mo	R ² , all folds	R ² , ≥ 18 mo
Naive: same hour D-1	21.7	22.3	0.42	0.46
Naive: same hour D-7	30.3	32.6	0.03	0.02
Ridge · + gas	23.0	17.7	0.39	0.71
Ridge · + gas + capacity	27.8	21.5	< 0 (diverges)	-2.38
LightGBM · weather + ENTSO-E forecasts	25.0	18.1	0.23	0.69
LightGBM · + gas	25.4	18.9	0.22	0.67
LightGBM · full, national weather	22.0	17.2	0.43	0.72
LightGBM · full, L1 loss	21.1	16.9	0.47	0.72
LightGBM · full + nuclear output D-2	20.3	16.6	0.52	0.74
LightGBM · full + all nuclear features	20.8	16.6	0.49	0.74
Hurdle + negative-price features	20.4	16.3	0.50	0.74
LightGBM · + gas + capacity	20.5	16.5	0.50	0.74
LightGBM · + neighbour prices*	11.4	11.4	0.79	0.83

EUR/MWh. R² below zero means a few extreme errors dominate: the linear model extrapolates on the trending capacity features.

Key findings

- **The model beats persistence by 26% once it has seen a full seasonal cycle:** MAE 16.5 vs 22.3 EUR/MWh, R² 0.74, on every one of the last 3 folds. With 6 to 13 months of history it does *worse* than persistence (Figure 2), which is why the 5-fold average gain is only 5%. In practice it needs at least 18 months of training data.
- **Installed capacity is the most valuable block.** Capacity-weighted renewable potential cuts the MAE from 18.9 to 16.5 on mature folds and stabilises the short-history folds (from 25.4 to 20.5 over all folds). Département-level weather beats national averages (17.2 $\rightarrow$ 16.5).
- **What did not help:** the 2-day-lagged gas price (18.1 without vs 18.9 with) and an L1 loss (16.9). A Ridge on weather and ENTSO-E forecasts is a strong linear baseline (17.7); LightGBM's edge comes from using the capacity features, which make the linear model unstable.
- **Nuclear availability adds nothing over yesterday's price (2024-2026).** Nuclear output of D-2, planned outages, their day-on-day change and the residual load left to thermal plants (16.63 / 16.63 vs 16.48). The model uses the thermal residual load (15% of gain) but as a substitute for residual load: the fleet moves over weeks, so its state is already priced in D-1. Caveat: ENTSO-E only serves the *last* revision of each REMIT message, republished in Oct. 2025, so forced outages were excluded to avoid look-ahead.
- **Neighbour prices are a leakage trap.** Adding them drops the MAE to 11.4 and a single coupled neighbour (Belgium) takes most of the model's importance. Results that include such features overstate what can be

achieved before the auction. This project's own first version made that mistake.

- **Negative prices remain the main blind spot.** Hours at or below 0 (mostly 10:00–16:00, weekends, April–June; 1.5% of hours in H1 2023, 11% in H1 2025) are over-forecast by +12 EUR/MWh on average, worse than persistence on those hours (MAE 18.0 vs 17.5). A two-regime *hurdle* model (classifier $P(\text{price} \leq 0) \times$ regime-specific regressors) with targeted features (renewable share, bridge days, solar on non-working days, negative hours at D-1/D-7) barely moves the needle: overall MAE 16.3, negative-hour MAE 17.7, bias +13. The classifier is never confident enough to switch regime, and the base rate keeps rising with solar capacity, which tree models cannot extrapolate.

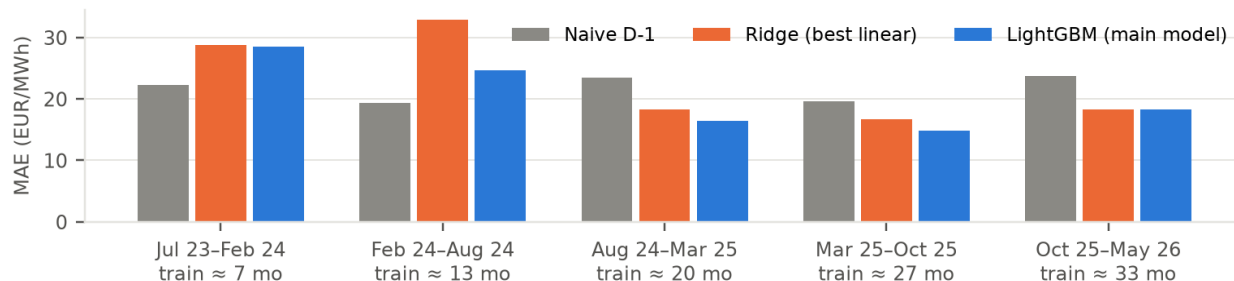


Figure 2 – MAE per test fold, with the length of the training window. The model needs a full seasonal cycle; once it has one, it beats persistence on every fold.

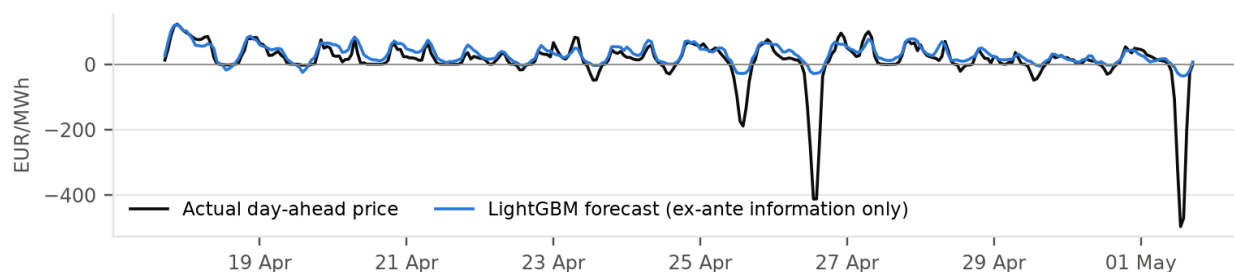


Figure 3 – Hourly forecast vs actual over the most volatile two weeks of the last test fold (17 Apr 2026 – 01 May 2026), out-of-sample. The level and daily shape are tracked; the negative-price troughs on sunny, windy weekends are missed.

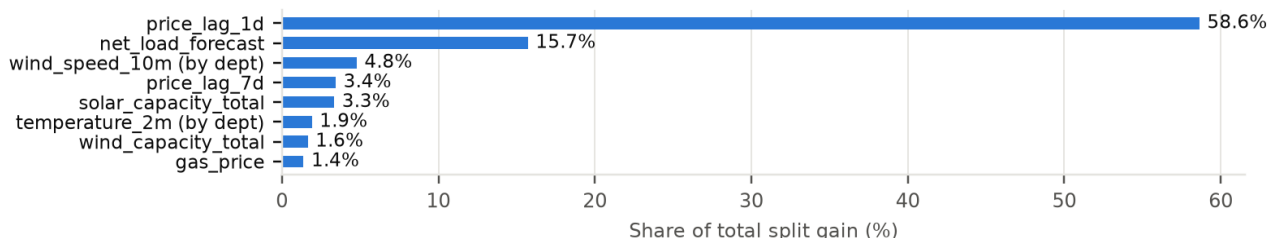


Figure 4 – Feature importance of the main model (share of split gain; département weather grouped by variable). Yesterday's price (59%) and the residual-load forecast (16%) dominate, followed by département wind speed and installed capacity.

5 · Limitations and next steps

- **Weather is reanalysis, not forecast.** Historical weather is ERA5 (what actually happened), whereas a live model sees D-1 weather forecasts, so backtest errors are slightly optimistic. Next step: train on archived forecast runs (Open-Meteo historical forecasts, ECMWF).
- **Point-in-time fundamentals:** nuclear features need a REMIT feed with the full revision history (as-of D-1 view, including forced outages); hydro reservoir levels, interconnection capacities and the EUA carbon price are still missing. They are the likeliest sources of the missed spikes.
- **From point forecast to P&L:** quantile forecasts (quantile LightGBM, or the Temporal Fusion Transformer already in the code) scored with pinball loss, so that the negative-price risk is a probability rather than a missed point; then a battery-arbitrage backtest at 15-minute resolution to express the gain in EUR.

Stack: Python, pandas, LightGBM, scikit-learn, PyTorch Forecasting (TFT), entsoe-py, Open-Meteo API. **Reproduce:** python experiments/run_ablations.py then python reports/build_report.py. **Code:** github.com/leomariocode/EnergeticPricing